

Social Media Interaction Analysis

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April 15, 2026

1 Introduction

Over the course of a year, a cosmetics company made social media posts on Facebook. Each Facebook post generated interactions defined as the sum of likes, comments, and shares. The company believes that higher engagement is strongly associated with higher short-term product demand. Posts are characterized by features available at the time of publication, including temporal variables (hour and weekday) and engagement metrics such as total and lifetime interactions from users, consumers, and followers. By accurately forecasting post interactions, the company can better prepare for fluctuations in order to improve operational planning.

This study will develop predictive models for social media interactions in two feature sets: a reduced and full model. Both will include features available at the time of posting but which differ by the inclusion of the Facebook post impressions, the number of times a post from a page was displayed whether clicked or not. Multiple linear regression will be used as a baseline model, along with regression techniques, to improve model performance and address skewness. Model performance is evaluated using root mean squared error (RMSE) which is carried out through cross-validation on a training/test data split. Furthermore, Ridge and Lasso regressions are applied to reduce potential overfitting and improve generalization.

2 Data Cleaning

This study uses the “Facebook metrics” dataset organized by Sergio Moro, Paulo, Rita, and Bernardo Vala for a research journal on predicting social media performance with brand building impacts ([1]). The original collection of 790 Facebook posts was gathered from January to December 2014 where it was filtered and cleaned into 500 social media posts with 19 features (variables). These social media posts are organized by month, weekday, and hour of day in a descending order from December to January with the following characteristics. For the purposes of this report, I renamed each of the features for organization purposes shown below.

Features Names:

- “Pages total likes” = “LikeTotal”
- “Category” = “Category”
- “Post Weekday” = “Weekday”
- “Paid” = “Paid”
- “Lifetime Post Total Impressions” = “Impressions”
- “Lifetime Post Consumers” = “Consumers”
- “Lifetime Post Impressions by people who have liked your Page” = “ImpressionsLiked”
- “Type” = “Type”
- “Post Month” = “Month”
- “Post Hour” = “Hour”
- “Lifetime Post Total Reach” = “Reach”
- “Lifetime Engaged Users” = “EngagedUsers”
- “Lifetime Post Consumptions” = “Consumptions”

- “Lifetime Post reach by people who like your Page” = “ReachedLiked”
- “Lifetime People who have liked your Page and engaged with your post” = “Engaged-Liked”
- “comment” = “Comments”
- “like” = “Likes”
- “share” = “Shares”
- “Total Interactions” = “Interactions”

Before loading the data, I uploaded this 500 observation dataset into Microsoft Excel and noticed that the features were contained in a single column and separated by a semicolon (;). To fix this, I selected the whole column, went to the Data tab and clicked “Text to Columns” with the “Delimited” then “Semicolon” in order to separate the data into their respective columns. Afterwards, I reloaded the dataset into a csv file to further check any missing variables using RStudio. To check any missing variables, a frequency table was created to check if there were any and five observations contained missing values in the “Shares” and “Interactions” columns. Using the `table()` function and `is.na()` function, the 5 observations containing the N/A values by a logical vector was used to remove them and now the dataset was left with 495 usable observations. From there I would model with the Interactions as my response variable with the following features at the time of posting with their respective characteristic levels.

Features:

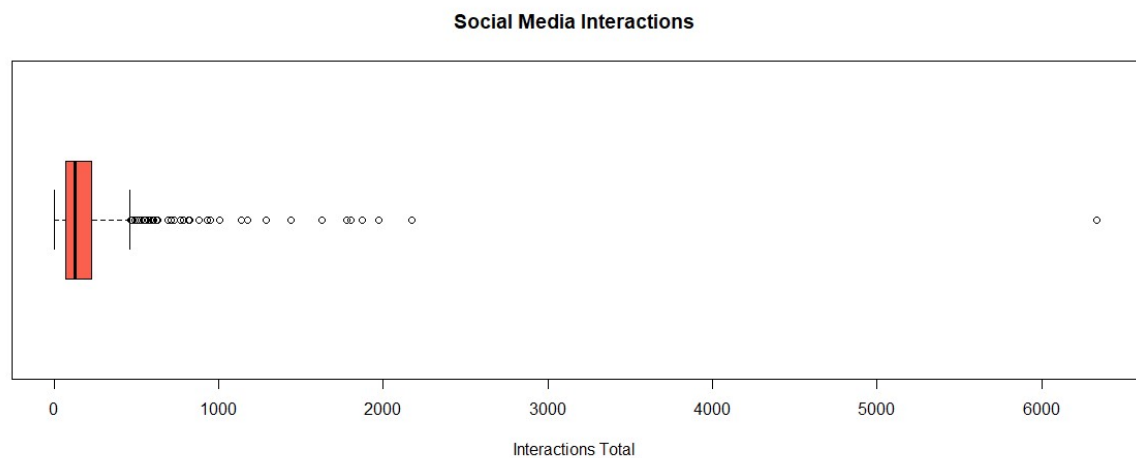
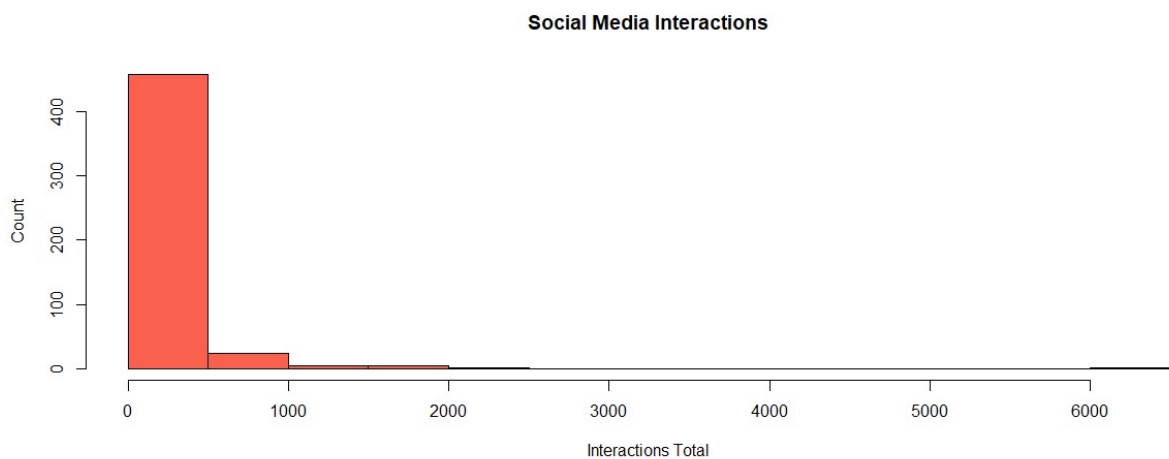
- *Interactions*: Quantitative response variable; the sum of likes, comments and shares
- *Category*: Type of content, classified as Action (special offers/contests), Product (direct advertising content), or Inspiration (non-promotional content).
- *Type*: Format of the post, including Link, Photo, Status, or Video.
- *Like Total*: Total number of page likes associated with the post.
- *Hour*: Hour of publication recorded on a 24-hour scale from 0 (12 am) to 23 (11 pm).
- *Weekday*: Day of the week when the post was published, coded as 1 (Sunday) to 7 (Saturday).
- *Paid*: Binary indicator of whether the post was promoted through paid advertising (0 = No, 1 = Yes).
- *Impressions*: Number of times a post was displayed to users, regardless of interaction.

3 Exploratory Data Analysis

For the response variable, Interactions have the following summary statistics. The clean data has a mean of 214 interactions and a standard deviation of 381.68 interactions. However table shows a maximum of 6,334 interactions. This would appear to be a huge outlier and later looking at the histogram and boxplot of the response variable it appears there are additional outliers beyond the extreme maximum. This extreme was outside the typical cluster of outliers from the 500 to 2000 range.

Table 1: Interactions Summary Statistics

Response Variable	Min	1st Qu.	Median	3rd Qu.	Max	Mean	SD
Interactions	0	72	125	231	6334	214	381.68



For its graphical depictions, the Facebook interactions is skewed to the right where a large number of outliers are outside of the upper Interquartile Range (IQR) which is around a sum of 500 interactions count. In order to fix this later in the modeling section, a logarithmic transformation will be used in order to fix this skewed relationship. No outliers were removed in order to preserve the distribution to track the number interactions that are associated to higher order placements. These outliers will be useful in determining this goal.

As for the features, the categorical variables each have their own respective classifications shown below with respective counts from the 495 observations.

Table 2: Categorical Feature Classifications

Variable	Definition	Variable	Definition
Category	1 = Action	Weekday	1 = Su (Sunday)
	2 = Product		2 = Mo (Monday)
	3 = Inspiration		3 = Tu (Tuesday)
Type	1 = Link		4 = We (Wednesday)
	2 = Photo		5 = Th (Thursday)
	3 = Status		6 = Fr (Friday)
	4 = Video		7 = Sa (Saturday)
Paid	0 = No (Not Paid/Free)	Hour	am = 0–11 = 12am–11am
	1 = Yes (Paid)		pm = 12–23 = 12pm–11pm

Category		Type		Paid	
Category	Count	Type	Count	Paid	Count
Action	211	Link	22	No	356
Product	129	Photo	421	Yes	139
Inspiration	155	Status	45	Total	495
Total	495	Video	7		
		Total	495		

Table 3: Weekday

Weekday	Su	Mo	Tu	We	Th	Fr	Sa	Total
Count	68	66	64	71	66	80	80	495

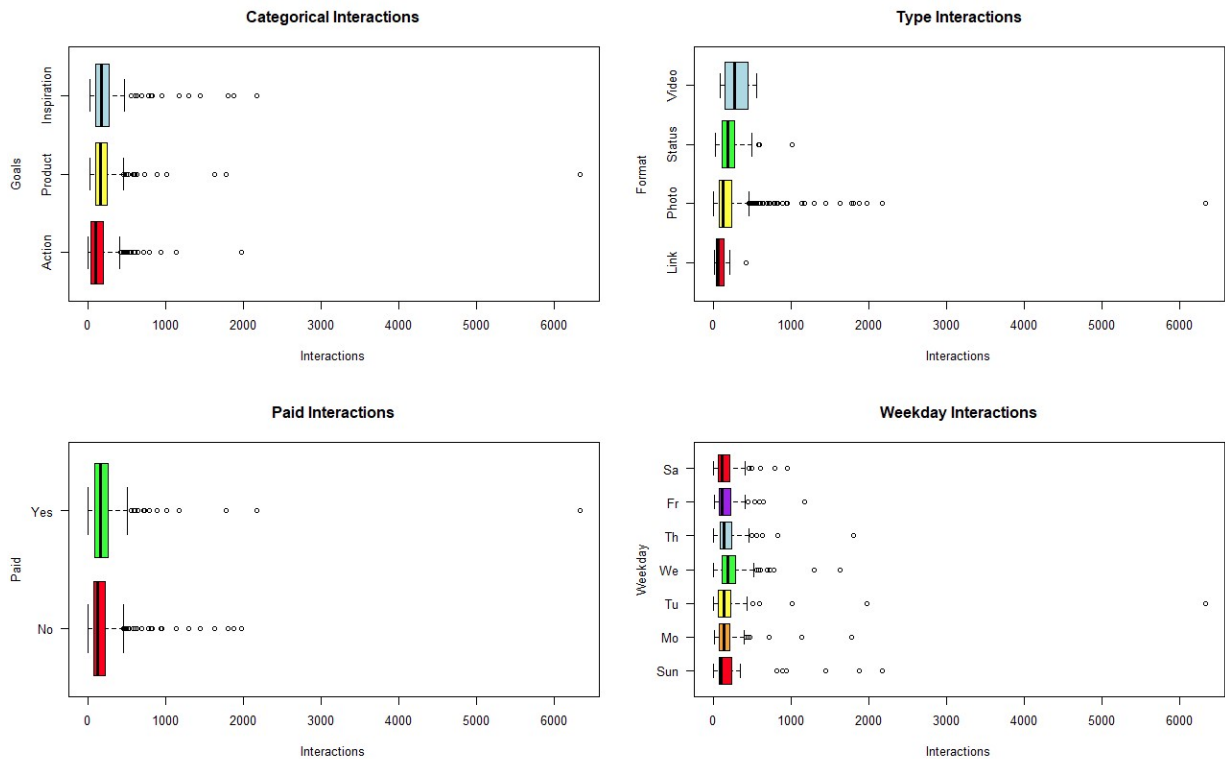
Table 4: Hour (am)

am	12	1	2	3	4	5	6	7	8	9	10	11	Total
Count	0	4	39	105	34	13	15	13	11	29	77	44	384

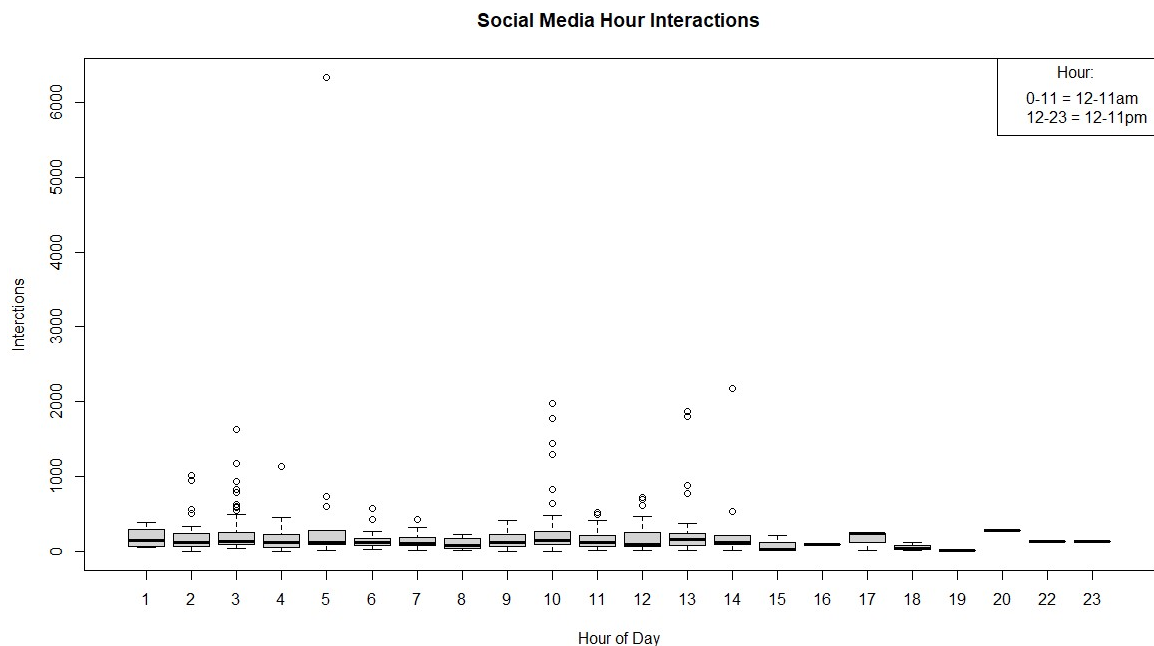
Table 5: Hour (pm)

pm	12	1	2	3	4	5	6	7	8	9	10	11	Total
Count	29	52	13	6	1	3	3	1	1	0	1	1	111

Looking at the categorical counts, the cosmetics company has made more Action posts than Product and Inspiration posts. The company appears to emphasize special offers and contests competitions more frequently than other content categories. The company was also producing more photo content rather than short status, link, or video posts in order to advertise their product offers to see their products or be engaged. In addition, most of the content was non-paid where the company didn't pay Facebook for their posts. About 28% of the 495 posts shows the company did pay but since most of the company's posts were special offers advertised in photo posts, it would make sense that there weren't many paid ads related to it.



In terms of their distributions, these posts were relatively evenly distributed throughout the week. There were 80 social media posts for both Friday and Saturday which would make sense since most shoppers would most likely shop on the weekends depending on the daily activities. Many of the Facebook posts happened during the morning spreading to a little past noon at 2 pm. There were 105 posts made during 3 am and then it gets spread out during the morning until there were spikes of social media activities around 10 am to 1 pm.

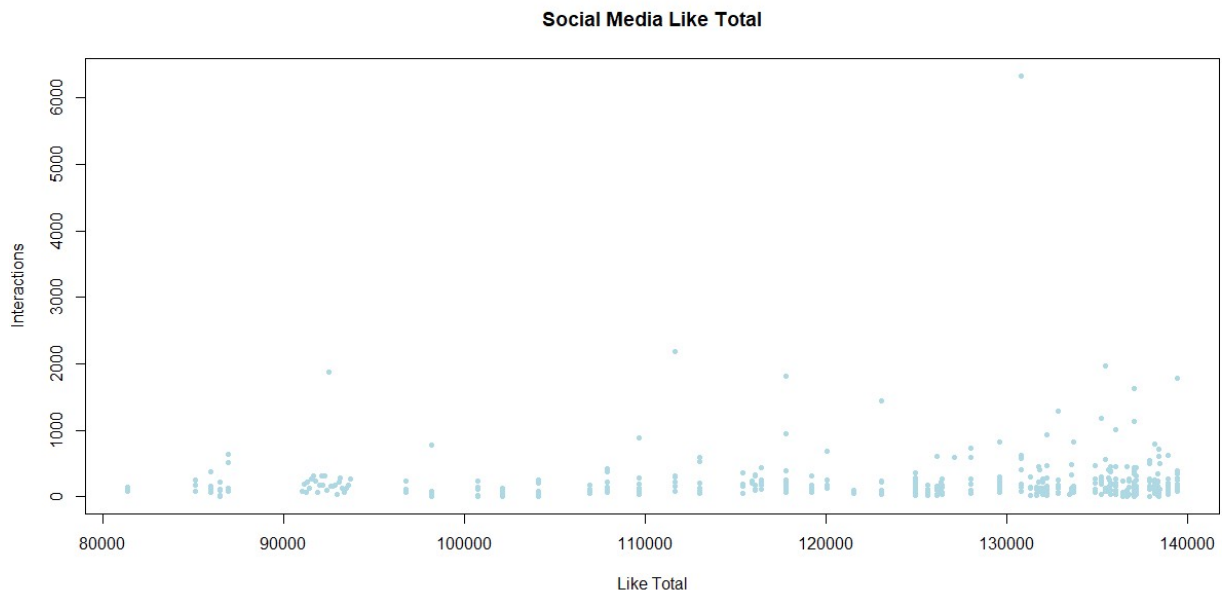


Also when comparing their distributions to the number interactions, they each show the same right skewed relationship to the Interactions. Each of the categorical features have a heavy upper tail where there are outliers clustered around the higher end of interactions. An example would be for the Type Interactions plot showing a cluster of points for the Photo format where there are two clusters of interactions with numbers around 500 to 1,000 and 1,000 to 2,000. For the maximum value of 6334 interactions, that observation has a few characteristics shown by the categorical features. This outlier appears to be a Product Photo post which is a Paid content made during 6 am on a Tuesday.

Table 6: Summary Statistics for Quantitative Features

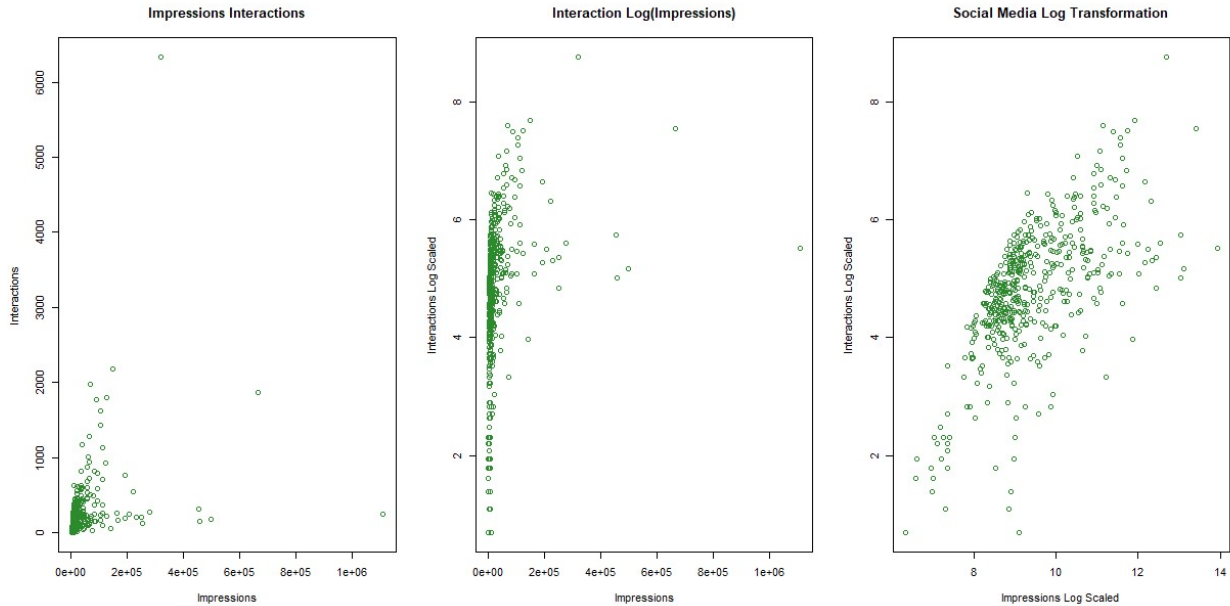
Variable	Min	1st Qu.	Median	3rd Qu.	Max	Mean	SD
LikeTotal	81370	112324	129600	136393	139441	123173	16203.82
Impressions	570	5798	9084	22503	1110282	29857	77142.91

For the quantitative (numerical) features, the Like Total and Impressions, their summary statistics are shown above. Like Total represents the total page likes at time of post. From all the Facebook posts, there was a mean of 123,173 likes with a standard deviation of about 16,204 likes. For its distribution, the relationship between LikeTotal and Interactions appears approximately linear where the values are concentrated between the first and third quartiles. No transformation was applied based on its linearity. A transformation for a feature when comparing it to the response variable will only occur if the relationship appears nonlinear or there are a cluster of points revealed in its distribution. Since there is none, the Like Total feature will be retained in its original form for modeling.



For the Impressions distribution, a transformation is needed. Impressions represents the number of times a post is displayed whether clicked or not. When looking at its statistics, It has a mean of approximately 29,857 impressions with a very large standard deviation. that's twice a large than the mean compared to the LikeTotal. There's a large gap between the third quartile and the maximum value which suggests a skewed distribution with outliers.

When comparing it to the Interactions, there are a cluster of points near the origin in the 1x3 table shown on the next page. In order to fix this, a transformation on the response variable was used to improve the nonlinear relationship. A log transformation was applied to the response variable but after transforming it (middle plot) the depiction still remains nonlinear so another transformation is applied to the Impressions feature to further reduce skewness. The final resulting plot would be the third plot on the 1x3 table on the right. Based on these graphical depictions, for the model selection a logarithmic transformation will be applied to the Impressions feature when selecting the right model.



4 Model Selection

Prior to my model selection, a train and test data split will be conducted. In order to accurately predict the number of interactions to prepare for future order volumes, a train and test split is appropriate to improve predictive accuracy and prevent overfitting. The data is split into two datasets: test and training set where the test dataset contains 20% of the dataset and the training dataset has the remaining 80%. For this dataset, the test data set will have 95 Facebook posts from the months November and December and the training dataset contains the remaining 400 posts. As for the two models, both will use the Interactions as the response variable but the Reduced model will be without the Impressions feature whereas the Full model has the Impressions feature. A multiple linear regression (MLR) base model shown below with the features descriptions will be used in Rstudio to find the best model for both models.

$$Y_i = \beta_0 + \beta_1 x_{i,\text{cat}} + \beta_2 x_{i,\text{type}} + \beta_3 x_{i,\text{paid}} + \beta_4 x_{i,\text{wday}} + \beta_5 x_{i,\text{hr}} + \beta_6 x_{i,\text{liketot}} + \beta_7 x_{i,\text{impr}} + \epsilon_i,$$

where

- Y = is the number of interactions; the sum of likes, comments and shares of post i .

- $x_{i,\text{cat}}$ is the category characterization for the post i .
- $x_{i,\text{type}}$ is the content type for the post i .
- $x_{i,\text{paid}}$ is the paid indicator variable for the post i .
- $x_{i,\text{wday}}$ is the weekday when the post i was published.
- $x_{i,\text{hr}}$ is hour the post i when published.
- $x_{i,\text{liketot}}$ is the total number of likes post i received.
- $x_{i,\text{impr}}$ is the total number impressions post i received (Full Model only).
- ϵ_i is the error of post i of the model

Both parts will include a MLR base model as well as other regression methods such as Ridge and Lasso Regressions. The minor difference is the Impressions feature exclusion in the Reduced model to the inclusion in the full model.

4.1 Reduced Model (Without Impressions)

For the Reduced model, a log transformation on the Interactions response variable was strategically applied. Since there can be some social media posts that may not get any interactions they may end up with 0 interactions. A $\log(0)$ output would tell Rstudio the output as $-\text{inf}$ (negative infinity). To compensate this, the response variable will be transformed as: $\log(\text{Interactions} + 1)$ where $\log(1) = 0$ in mathematical terms. As a result the reduced MLR model was transform into a log-linear regression model as follows:

$$\log(Y_i + 1) = \beta_0 + \beta_1 x_{i,\text{cat}} + \beta_2 x_{i,\text{type}} + \beta_3 x_{i,\text{paid}} + \beta_4 x_{i,\text{wday}} + \beta_5 x_{i,\text{hr}} + \beta_6 x_{i,\text{liketot}} + \epsilon_i,$$

where

- $\log(Y_i + 1)$ = is the number of interactions plus 1; the sum of likes, comments and shares of post i plus 1, transformed under the logarithmic scale.

Using these features, Candidate models containing polynomial terms were evaluated using Akaike Information Criterion (AIC). This metric was used to see which model was better and the lower the metric, the better. When comparing it to the other models with nonlinear terms, the log-linear model had the best predictive accuracy of 1221 compared to the based model AIC of 5953. The log-linear model substantially improved predictive performance. This model was chosen with its statistics shown next page.

Table 7: AIC Reduced Model

Model	AIC
Reduced MLR	5953
Log MLR	1221

Table 8: Multiple Linear Regression Model (Reduced Model)

Predictor	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.660e+00	5.166e-01	7.084	6.55e-12
Category	4.643e-01	6.614e-02	7.019	9.92e-12
TypePhoto	3.673e-01	3.002e-01	1.223	0.22197
TypeStatus	6.582e-01	3.627e-01	1.815	0.07035
TypeVideo	1.906e+00	7.057e-01	2.701	0.00722
Paid	1.130e-01	1.221e-01	0.925	0.35561
Weekday	-6.267e-02	2.727e-02	-2.298	0.02207
Hour	-1.888e-02	1.266e-02	-1.491	0.13677
LikeTotal	1.680e-06	3.571e-06	0.470	0.63828

From this newly transformed model, the MLR summary statistics reveal the β_0 intercept value is significant and the same can be said for the Category and Weekday feature. Both have a p-value that is less than the $\alpha = 0.05$ level what's most surprising is that even after using the `as.factor()` function in RStudio, the Type expanded into dummy variables. This is done since Type is a categorical feature with levels as either: Link, Photo, Status or Video. For their level significance, Video is the only significant one because viewing the EDA analysis on page 6 the light blue boxplot of the Video Type format shows no outliers. This suggests that video posts may be associated with higher engagement levels unlike Like Total which is non significant.

K-fold cross-validation (with $k = 10$) was used to obtain a more robust estimate of model performance. The cross-validated RMSE (1.12) is slightly higher than the training RMSE (1.09), which is expected due to reduced overfitting in cross-validation. These values are reported on the log-transformed response scale, so differences reflect predictive error in log-units rather than raw interaction counts.

Table 9: RMSE Comparison (Reduced model)

Model	RMSE
Reduced Model	1.09
k-fold	1.12

From here, the regression methods such as Ridge and Lasso regression were conducted. Both regression techniques were also used to improve the log-linear model's accuracy with the inclusion of penalty terms. These penalty parameters acted as a limiter to strengthen the model but the difference is Ridge regression shrinks coefficients toward zero to reduce variance, while Lasso regression can shrink some coefficients exactly to zero. In order to set up this model, a number of things are needed a Y vector and X matrix is created from the MLR model where its rewritten as:

$$\mathbf{Y} = \mathbf{X}\beta + \epsilon, \text{ where}$$

$$\mathbf{Y} = \begin{bmatrix} \log(Y_1 + 1) \\ \log(Y_2 + 1) \\ \vdots \\ \log(Y_{400} + 1) \end{bmatrix}, \quad \beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_7 \end{bmatrix}, \quad \epsilon = \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \epsilon_3 \\ \vdots \\ \epsilon_{400} \end{bmatrix}$$

$$\mathbf{X} = \begin{bmatrix} 1 & x_{1,\text{cat}} & x_{1,\text{type}} & x_{1,\text{paid}} & x_{1,\text{wday}} & x_{1,\text{hr}} & x_{1,\text{liketot}} \\ 1 & x_{2,\text{cat}} & x_{2,\text{type}} & x_{2,\text{paid}} & x_{2,\text{wday}} & x_{2,\text{hr}} & x_{2,\text{liketot}} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_{400,\text{cat}} & x_{400,\text{type}} & x_{400,\text{paid}} & x_{400,\text{wday}} & x_{400,\text{hr}} & x_{400,\text{liketot}} \end{bmatrix}$$

Using the X matrix and Y vector, a grid of 100 penalization parameters from 2^{-10} to 2^{10} was created to be applied with the “glmnet” R package containing Ridge and Lasso Regression. By the `cv.glmnet()` function and the alpha setting where alpha = 1 for Lasso and alpha = 0 for Ridge, cross-validation was formally conducted which gives an approximation of smallest lambda value and RMSE for both methods. The minimum lambda values were about $4.3 * 10^{-2}$ for Lasso and about $1.1 * 10^{-1}$ for Ridge. For the Lasso terms that were shrunk to zero, They consist of the categorical levels of Type (Photo & Status), Paid, Weekday (Mo, Tue, & Th), Hour (morning & afternoon) and LikeTotal. None of the Ridge terms shrunk to zero but they were approaching it by their scientific notation.

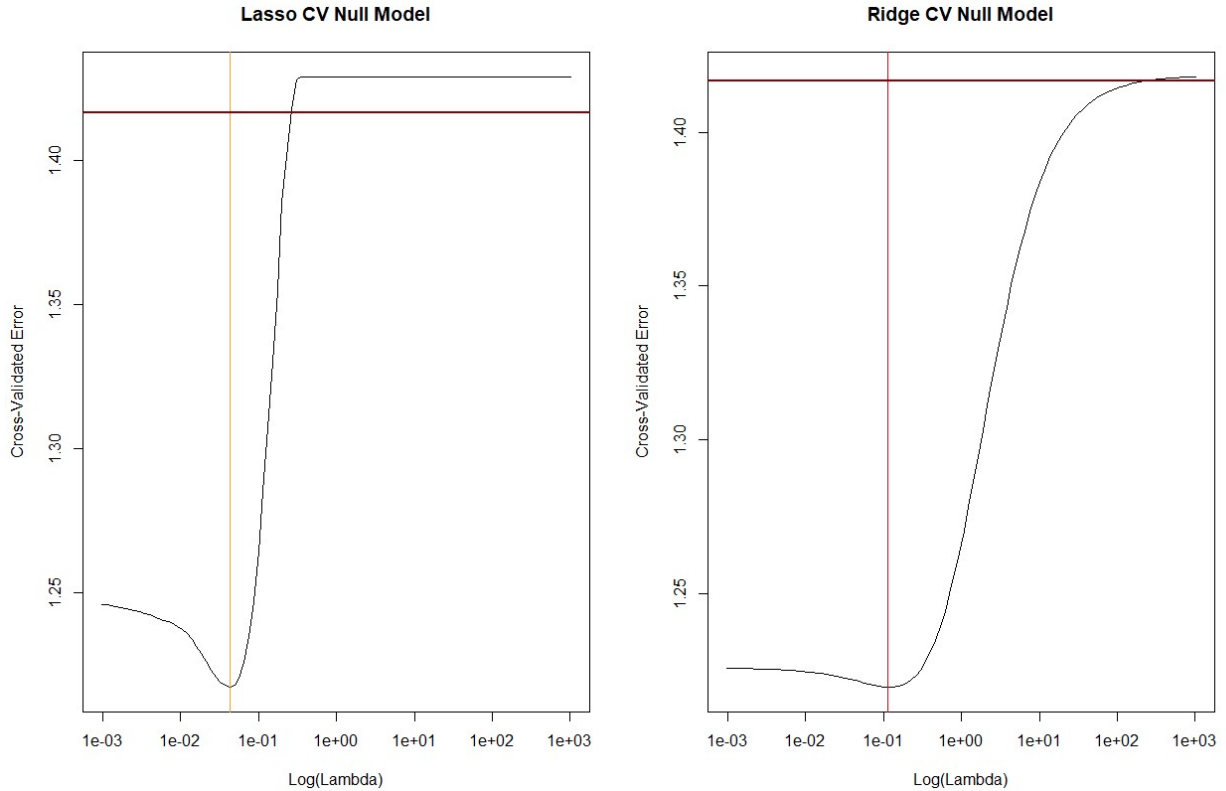


Table 10: Minimum Lambda and RMSE (Reduced Model)

Model	Minimum Lambda	RMSE
Lasso	0.04282	1.069
Ridge	0.1141	1.060

From the log lambdas vs error plots, both depictions had a huge dip that reveals the minimum lambda values for Lasso and Ridge. A brown horizontal line represents baseline model error was included to see how well the errors were contained. Lasso flattens out but is not as well contained by the baseline error as shown by the area above the baseline error. Ridge on the other hand handled it better despite some area not contained. While the minimum lambda values were different for both regression methods so are the RMSE values. Lasso has an RMSE of about 1.07 whereas Ridge has a 1.06 RMSE metric. Both are slight improvements compared to the k-fold Reduced MLR model RMSE of 1.12 but Lasso appears to be the superior RMSE since it has the smallest RMSE, similar to the AIC principles.

4.2 Full Model (With Impressions)

Similarly to the Reduced model, a log transformation on the Interactions response variable was also applied. However, the inclusion of the Impressions feature in the full model was a change for the better. The Impressions feature was more valuable as shown in the correlation matrix in the tables appendix (page 17). Impressions improved interactions. Also since transformations were used on the Impressions feature, the feature will be transformed as: $\log(\text{Impressions})$. As a result the base MLR full model with the inclusion of the Impressions feature has the following model:

$$\log(Y_i + 1) = \beta_0 + \beta_1 x_{i,\text{cat}} + \beta_2 x_{i,\text{type}} + \beta_3 x_{i,\text{paid}} + \beta_4 x_{i,\text{wday}} + \beta_5 x_{i,\text{hr}} + \beta_6 x_{i,\text{liketot}} + \beta_7 \log(x_{i,\text{impr}}) + \epsilon_i,$$

where

- $\log(Y_i + 1)$ = is the number of interactions plus 1; the sum of likes, comments and shares of post i plus 1 transformed under the logarithmic scale.
- $\log(x_{i,\text{impr}})$ is the total number of impressions post i received, transformed under the logarithmic scale.

Some trial and error with the feature values was also carried out by adding/removing nonlinear terms to this model with LikeTotal and $\log(\text{Impressions})$ features, but I ultimately chose to leave the model as shown above with the following estimates. This is because the nonlinear features were nonsignificant despite the AIC slightly improving. The AIC for this full model, post transforming on the Interactions and Impressions, gives an AIC of about 943. This shows a huge improvement compared to the base model and the log-linear Reduced model. Also, the full model has more statistically significant features, indicating an improvement in model fit. The LikeTotal and the $\log(\text{Impressions})$ features have a p-value that is less than the $\alpha = 0.05$ level which includes the intercept and the Type feature levels. By further testing it with k-fold cross-validation, it was conducted the log-linear full model

Table 11: Multiple Linear Regression Model (Full Model)

Predictor	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-4.055e+00	5.340e-01	-7.594	2.32e-13
Category	5.333e-01	4.689e-02	11.373	< 2e-16
TypePhoto	4.767e-01	2.124e-01	2.245	0.025337
TypeStatus	1.665e-01	2.577e-01	0.646	0.518596
TypeVideo	6.458e-01	5.030e-01	1.284	0.199925
Paid	-4.123e-02	8.671e-02	-0.475	0.634733
Weekday	-2.166e-02	1.939e-02	-1.117	0.264596
Hour	-1.039e-02	8.964e-03	-1.159	0.247203
LikeTotal	9.384e-06	2.555e-06	3.673	0.000273
log(Impressions)	6.828e-01	3.448e-02	19.803	< 2e-16

Table 12: AIC Full Model

Model	AIC
Reduced MLR	5900
Log MLR	943

Table 13: RMSE Comparison (Full Model)

Model	RMSE
Full Model	0.767
k-fold	0.792

got an improved RMSE of about 0.79. Without cross-validation RMSE is about 0.77 which is improved from the Reduced model's accuracy.

From this I went to predictive modeling with this version into the Ridge and Lasso regressions where the MLR model will be rewritten as follows with the Y and ϵ vectors stay the same.

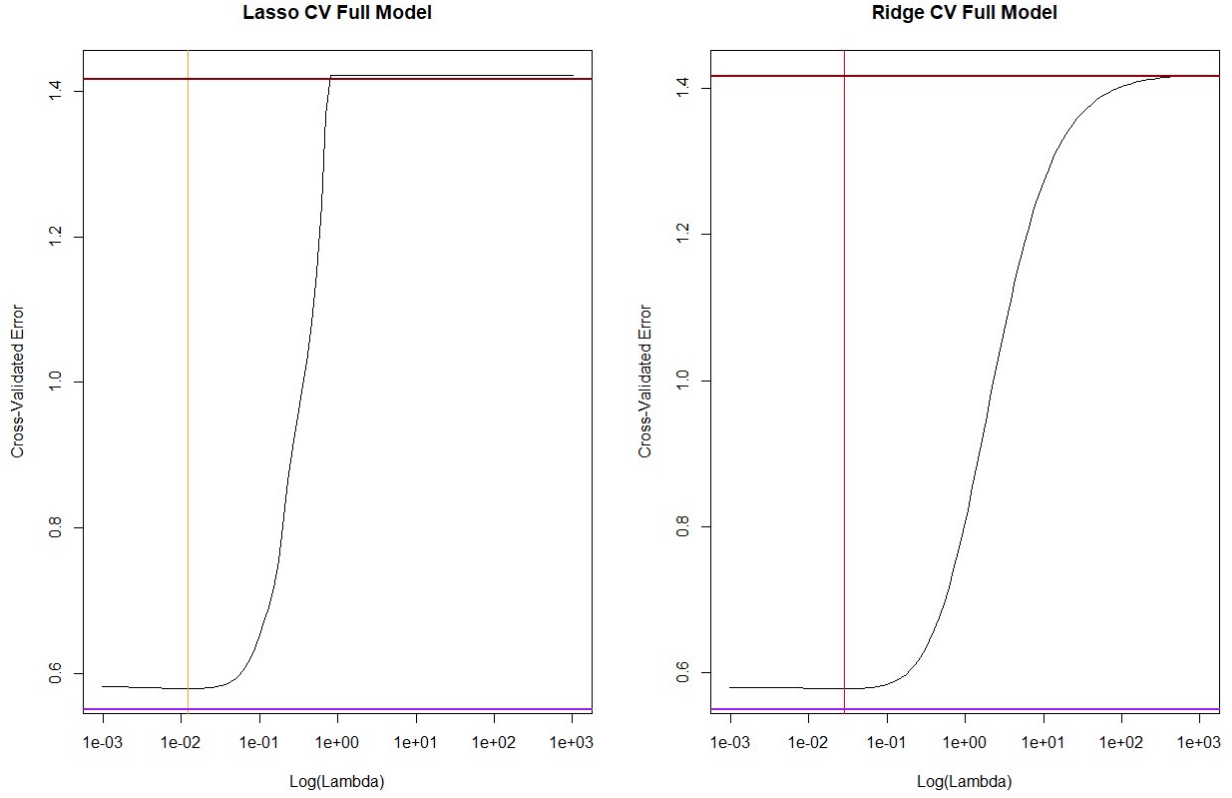
$$\mathbf{Y} = \mathbf{X}\beta + \epsilon, \text{ where}$$

$$\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_8 \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} 1 & x_{1,\text{cat}} & x_{1,\text{type}} & x_{1,\text{paid}} & x_{1,\text{wday}} & x_{1,\text{hr}} & x_{1,\text{liketot}} & \log(x_{1,\text{impr}}) \\ 1 & x_{2,\text{cat}} & x_{2,\text{type}} & x_{2,\text{paid}} & x_{2,\text{wday}} & x_{2,\text{hr}} & x_{2,\text{liketot}} & \log(x_{2,\text{impr}}) \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_{400,\text{cat}} & x_{400,\text{type}} & x_{400,\text{paid}} & x_{400,\text{wday}} & x_{400,\text{hr}} & x_{400,\text{liketot}} & \log(x_{400,\text{impr}}) \end{bmatrix}$$

Using the updated X matrix and Y vector with the same set of 100 penalized parameters, Lasso and Ridge regressions were obtained via cross-validation. This time, by the log scaled lambda with MSE values the approximations are much better in terms of the variance brown horizontal line. The error values are better contained in the Lasso and Ridge regression models depicted by the brown line. The purple line contains the MSE for the respective models and the minimum lambda increased from both the Lasso and Ridge models. With the inclusion of the log-scaled Impressions, Lasso had a minimum lambda of 1.2×10^{-2} whereas Ridge had 2.8×10^{-2} . For Lasso, the terms that were shrunk to zero were Weekday (Tuesday and Friday) and for Hour (Morning and Evening). Ridge also had some terms approaching zero but it wasn't a huge difference. The updated RMSE metrics for both models decreased to about 0.73 but Ridge is slightly better. This shows the inclusion of the logarithmic scaled Impressions feature improves the model's accuracy.

Table 14: Minimum Lambda and RMSE (Full Model)

Model	Minimum Lambda	RMSE
Lasso	0.01214	0.728
Ridge	0.02813	0.725



5 Result Comparison

After looking on the train data, the models were then evaluated using RMSE on the test dataset for comparison. For the MLR reduced and full models, I would use the RMSE formula,

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

to accurately predict the interactions for short term order on the 95 observations from the test dataset. Using the same models for the reduced and full model, I performed the k-fold cross-validation on the MLR model as well as Lasso and Ridge regression where $k = 10$. For Lasso and Ridge regression it required me to create a new Y vector and X matrix specifically tuned for the test data's 95 observations. The model is shown next page where the Reduced model, would be without the column containing the $\log(x_i, impr)$ for $i = 1 : 95$.

$$\mathbf{Y} = \mathbf{X}\beta + \epsilon, \quad \text{where}$$

$$\mathbf{Y} = \begin{bmatrix} \log(Y_1 + 1) \\ \log(Y_2 + 1) \\ \log(Y_3 + 1) \\ \vdots \\ \log(Y_{95} + 1) \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} 1 & x_{1,\text{cat}} & x_{1,\text{type}} & x_{1,\text{paid}} & x_{1,\text{wday}} & x_{1,\text{hr}} & x_{1,\text{liketot}} & \log(x_{1,\text{impr}}) \\ 1 & x_{2,\text{cat}} & x_{2,\text{type}} & x_{2,\text{paid}} & x_{2,\text{wday}} & x_{2,\text{hr}} & x_{2,\text{liketot}} & \log(x_{2,\text{impr}}) \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_{95,\text{cat}} & x_{95,\text{type}} & x_{95,\text{paid}} & x_{95,\text{wday}} & x_{95,\text{hr}} & x_{95,\text{liketot}} & \log(x_{95,\text{impr}}) \end{bmatrix}$$

After computing the RMSE for both regressions in the reduced and full model in the test dataset, the results look consistent to each other. The Reduced model RMSEs for Ridge and Lasso from the test data is slightly lower and has better performance. This is possibly from sample size reduction, but when compared with the RMSE of the train data, the k-fold RMSE = 1.12 and the Ridge/Lasso RMSE has about 1.065. The test RMSE for the MLR is 1.06 whereas Ridge and Lasso have 1.05. The full model achieves lower RMSE values compared to the reduced model, indicating improved predictive performance on the log-transformed scale. The test RMSE (0.74) suggest smaller average prediction error in log-interaction unites, consistent with improved generalization when Impressions is included. The train data RMSEs are lower for Ridge which appears to be the best RMSE compared to k-fold and Lasso which are higher than the 0.73 train metric.

RMSE Train (Reduced Model)

Model	RMSE
MLR	1.09
k-fold	1.12
Lasso	1.07
Ridge	1.06

RMSE Train (Full Model)

Model	RMSE
MLR	0.767
k-fold	0.791
Lasso	0.728
Ridge	0.725

RMSE Test (Reduced Model)

Model	RMSE
MLR	1.06
Lasso	1.05
Ridge	1.05

RMSE Test (Full Model)

Model	RMSE
MLR	0.745
Lasso	0.733
Ridge	0.741

6 Summary

In this report, I developed multiple predictive models for Facebook post interactions using both a reduced and a full model. Both models included a log-transformed response variable, Interactions, using $\log(\text{Interactions} + 1)$ to address right-skewness and improve model fit. Exploratory analysis showed that interactions were highly skewed with several extreme outliers. The Categorical features also exhibited clustering patterns and extreme-value behavior, including the observation with 6,334 interactions. From this analysis, the Impressions and Category appeared to be the most influential features whereas Weekday and Hour showed weaker effects to improve predictive accuracy.

Across the different multiple modeling approaches, the full model consistently outperformed the Reduced model in terms of RMSE. While the Reduced model improved with a log transformed Interactions, the inclusion of the Impressions feature improve the model's accuracy. Also transforming the Impressions feature using a logarithmic scale further improved the AIC confirms its importance as a key driver of engagement. Lasso and Ridge

regression continued to improved the full model's stability by reducing overfitting with an approximation of the minimum lambda value by cross-validation. The full model better encapsulates the penalty parameters with a lower RMSE of 0.73 compared to the Reduced model's 1.06 RMSE metric. Since the response variable is log-transformed, RMSE values were interpreted on the logarithmic scale.

Overall, Ridge regression on the full model provides the best predictive performance. Even with k-fold cross-validation where $k = 10$, Ridge regression remained the better approximation. While cross-validation slightly increased RMSE relative to the training fit, it provided a more reliable performance. If the cosmetics company would want to predict short-term orders, the full model under Ridge regression would be the best when viewing their Facebook interactions.

References

- [1] Sérgio Moro, Paulo Rita, and Bernardo Vala. Facebook metrics. UCI Machine Learning Repository, 2016.

7 Appendices

7.1 Tables

Table 15: Correlation Matrix of Quantitative Features

	Interactions	like_total	impression_total
Interactions	1.000	0.048	0.342
like_total	0.048	1.000	-0.103
impression_total	0.342	-0.103	1.000

Ridge and Lasso Coefficients (Reduced Model)

Lasso Coefficients		Ridge Coefficients	
Coefficient	Lasso	Coefficient	Ridge
Intercept	4.328	Intercept	4.028e+00
X1-Product	0.689	X1-Product	6.473e-01
X1-Inspiration	0.825	X1-Inspiration	8.234e-01
X2-Photo	.	X2-Photo	2.405e-01
X2-Status	.	X2-Status	3.511e-01
X2-Video	1.106	X2-Video	1.702e+00
X3	.	X3	4.381e-02
X4-Mo	.	X4-Mo	-7.065e-02
X4-Tu	.	X4-Tu	2.767e-02
X4-We	0.271	X4-We	3.159e-01
X4-Th	.	X4-Th	3.133e-02
X4-Fr	-0.024	X4-Fr	-1.243e-01
X4-Sa	-0.364	X4-Sa	-4.553e-01
X5-morning	.	X5-morning	-1.265e-01
X5-afternoon	.	X5-afternoon	-8.918e-02
X5-evening	-0.366	X5-evening	-6.829e-01
X6	.	X6	1.314e-06

Ridge and Lasso Coefficients (Full Model)

Lasso Coefficients		Ridge Coefficients	
Coefficient	Lasso	Coefficient	Ridge
Intercept	-3.318e+00	Intercept	-3.378e+00
X1-Product	1.017e+00	X1-Product	1.030e+00
X1-Inspiration	1.030e+00	X1-Inspiration	1.039e+00
X2-Photo	4.351e-01	X2-Photo	4.586e-01
X2-Status	-1.418e-01	X2-Status	-1.709e-01
X2-Video	6.208e-01	X2-Video	8.028e-01
X3	-3.477e-02	X3	-6.593e-02
X4-Mo	-7.066e-02	X4-Mo	-1.059e-01
X4-Tu	.	X4-Tu	-2.681e-02
X4-We	8.387e-02	X4-We	1.167e-01
X4-Th	6.465e-02	X4-Th	1.007e-01
X4-Fr	.	X4-Fr	1.768e-02
X4-Sa	-2.590e-01	X4-Sa	-2.838e-01
X5-morning	.	X5-morning	2.073e-03
X5-afternoon	-9.040e-02	X5-afternoon	-1.106e-01
X5-evening	.	X5-evening	5.537e-02
X6	6.131e-06	X6	6.719e-06
X7	6.840e-01	X7	6.813e-01

7.2 R Code

```

1
2 # Midterm project
3
4 # Step 1; LOAD IT!!!!
5 library(readr)
6 FB_Metrics <- read_csv("Spring 2026/STA 4990/Midterm Project/
   Facebook Metrics.csv")
7 View(FB_Metrics)
8
9 str(FB_Metrics)
10
11 attach(FB_Metrics)
12
13 #STEP 2; remove any missing values
14
15 # frequency table
16 lapply(FB_Metrics, table)
17 lapply(FB_Metrics, anyNA)
18 # share and total interactions
19
20 # check variables share and total interactions

```

```

21 # see how many missing values we have:
22 table(is.na(FB_Metrics$share))
23 table(is.na(FB_Metrics$'Total Interactions'))
24
25 # see if there is overlap in observations with missing values:
26 table(is.na(FB_Metrics$share),is.na(FB_Metrics$'Total Interactions'))
27
28 # create a logical vector indicating missingness for either variable
29 :
29 miss <- is.na(FB_Metrics$share) | is.na(FB_Metrics$'Total
    Interactions')
30
31 # look at the observations with any missing values to see if we can
    identify
32 # any pattern/explanation:
33 FB_Metrics[miss,]
34
35 # since we have a small number of missing values with no clear
    pattern/reason,
36 # we will simply remove the missing observations:
37 FB <- FB_Metrics[!miss,]
38 nrow(FB)
39 lapply(FB,anyNA)
40 View(FB)
41
42 detach(FB_Metrics)
43 attach(FB)
44 str(FB)
45
46 # STEP 2.5; Variable setup EDA ANALYSIS
47
48 # get variable names
49 names(FB)
50
51 # rename variables
52 names(FB) <- c(
53   "LikeTotal", "Type", "Category", "Month", "Weekday", "Hour", "Paid
    ",
54   "Reach", "Impressions", "EngagedUsers", "Consumers", "Consumptions
    ",
55   "ImpressionsLiked", "ReachLiked", "EngagedLiked",
56   "Comments", "Likes", "Shares", "Interactions"
57 )
58
59 # check to see name changes
60 View(FB)

```

```

61
62 # Y variable
63 Interactions <- FB$Interactions
64
65 # X's
66 LikeTotal <- FB$LikeTotal
67 Type <- FB$Type
68 Category <- FB$Category
69 Month <- FB$Month
70 Weekday <- FB$Weekday
71 Hour <- FB$Hour
72 Paid <- FB$Paid
73 Reach <- FB$Reach
74 ImpressionsTotal <- FB$Impressions
75 Engagement <- FB$EngagedUsers
76 Consumers <- FB$Consumers
77 Consumption <- FB$Consumptions
78 ImpressionLiked <- FB$ImpressionsLiked
79 LikeReach <- FB$ReachLiked
80 LikeEngaged <- FB$EngagedLiked
81 Comments <- FB$Comments
82 Likes <- FB$Likes
83 Shares <- FB$Shares
84
85 # need to use features when they make the post
86
87 # part 1; predictive modeling
88 Category;
89 Type;
90 LikeTotal;
91 Hour;
92 Weekday;
93 Paid;
94
95 # part 2;
96 ImpressionsTotal
97
98 # correlation matrix
99 with(FB, cor(data.frame("Interactions" = Interactions,
100                        "like_total" = LikeTotal,
101                        "impression_total" = ImpressionsTotal)))
102
103 # correlation scatter plot
104 with(FB, pairs(data.frame("Interactions" = Interactions,
105                          "like_total" = LikeTotal,
106                          "impression_total" = ImpressionsTotal) ) )
107

```

```

108 # categorical
109 Category <- as.factor(Category)
110 levels(Category) <- c("Action", "Product", "Inspiration" )
111 levels(Category)
112 Category
113
114 Type <- as.factor(Type)
115 levels(Type)
116 Type
117
118 Paid <- as.factor(Paid)
119 levels(Paid) <- c("No", "Yes")
120 levels(Paid)
121 Paid
122
123 Weekday <- as.factor(Weekday)
124 levels(Weekday) <- c("Sun", "Mo", "Tu", "We", "Th", "Fr", "Sa")
125 levels(Weekday)
126 Weekday
127
128 Hour <- as.factor(Hour)
129 levels(Hour)
130 Hour
131
132 # use categorical vars
133 Category;
134 Type;
135 Paid;
136 Weekday;
137 Hour;
138
139 # quantitative
140 LikeTotal;
141 ImpressionsTotal; # part 2
142
143 # STEP 3; EDA Analysis
144
145 # five number summary and mean of Interactions with histogram, box
    plot and
146 # stacked dot plot
147 summary(Interactions)
148 sd(Interactions)
149
150 par(mfrow=c(2,1))
151 hist(Interactions, col = "tomato", xlab = "Interactions Total",
    ylab = "Count",
152     main = "Social Media Interactions")

```

```

153 boxplot(Interactions, horizontal = TRUE, col = "tomato", xlab = "
      Interactions Total",
154         main = "Social Media Interactions")
155 stripchart(Interactions, method = "stack", xlab = "Interactions
      Total", ylab = "Count",
156            pch = 20, main = "Social Media Interactions")
157
158 # reset
159 par(mfrow=c(1,1))
160
161 # categorical
162 par(mfrow=c(2,2))
163 summary(Category)
164 plot(Category, Interactions, col = c("red", "yellow", "lightblue"),
165      xlab = "Interactions", ylab = "Goals",
166      main = "Categorical Interactions",
167      horizontal = TRUE )
168
169 summary(Type)
170 plot(Type, Interactions, col = c("red", "yellow", "green", "
      lightblue"),
171      xlab = "Interactions", ylab = "Format",
172      main = "Type Interactions",
173      horizontal = TRUE)
174
175 summary(Paid)
176 plot(Paid, Interactions, col = c("red", "green"),
177      xlab = "Interactions", ylab = "Paid", main = "Paid
      Interactions",
178      horizontal = TRUE, las = 1)
179
180 summary(Weekday)
181 plot(Weekday, Interactions,
182      col = c("red", "orange", "yellow", "green", "lightblue", "
      purple"),
183      xlab = "Interactions", ylab = "Weekday", main = "Weekday
      Interactions",
184      horizontal = TRUE, las = 1)
185
186 # reset
187 par(mfrow=c(1,1))
188 summary(Hour)
189 plot(Hour, Interactions, xlab = "Hour of Day", ylab = "Interctions
      ",
190      main = "Social Media Hour Interactions")
191 legend("topright",
192      legend = c("0-11 = 12-11am", "12-23 = 12-11pm"),

```

```

193         title = "Hour:")
194
195     # Quantitative
196     summary(LikeTotal)
197     sd(LikeTotal)
198     plot(LikeTotal, Interactions, col = "lightblue", pch = 20, xlab =
199           "Like Total",
200           ylab = "Interactions", main = "Social Media Like Total")
201     # looks fine
202
203     par(mfrow=c(1,3))
204     summary(ImpressionsTotal)
205     sd(ImpressionsTotal)
206     plot(ImpressionsTotal, Interactions, col = "forestgreen", pch =
207           21,
208           xlab = "Impressions", ylab = "Interactions",
209           main = "Impressions Interactions")
210
211     # needs transformations
212     plot(ImpressionsTotal, log(Interactions), xlab = "Impressions",
213           col = "forestgreen", pch = 21,
214           ylab = "Interactions Log Scaled",
215           main = "Interaction Log(Impressions)")
216     # still rough transform x
217
218     plot(log(ImpressionsTotal), log(Interactions), xlab = "Impressions
219           Log Scaled",
220           col = "forestgreen", pch = 21,
221           ylab = "Interactions Log Scaled",
222           main = "Social Media Log Transformation")
223     # 'Saul Good man :)
224
225     par(mfrow=c(1,1))
226
227     # STEP 4; hold 95 observations from last 2 months (11 and 12)
228
229     set.seed(4990)
230     test <- FB[1:95, ] # first 95, 11 and 12 months
231     train <- FB[96:495, ] # last 400; 1 - 10 months
232
233     nrow(test) # 95
234     nrow(train) # 400
235
236     # check to see if names are good
237     names(test)
238     names(train)

```

```

237 names(test) == names(FB)
238 names(train) == names(train)
239 names(test) == names(train)
240
241 detach(FB)
242
243 ##### STEP 5; MODEL SELECTION(MLR, ridge and/or lasso) ====
244
245 # variables to use =====
246 # use categorical vars
247 Category;
248 Type;
249 Paid;
250 Weekday;
251 Hour;
252
253 # quantitative
254 LikeTotal;
255 ImpressionsTotal; # part 2, renamed
256
257 # MLR part 1 =====
258 mlr1 <- lm(Interactions ~ Category + Type + Paid + Weekday + Hour
259           + LikeTotal, data = train)
260 summary(mlr1)
261
262 mlr12 <- lm( log(Interactions + 1) ~ Category + Type + Paid +
263            Weekday + Hour
264            + LikeTotal, data = train)
265 summary(mlr12)
266 # log transformation; add 1 since some had interactions = 0
267
268 mlr13 <- lm( log(Interactions + 1) ~ Category + Type + Paid +
269            Weekday + Hour
270            + LikeTotal + I(LikeTotal^2) , data = train)
271 summary(mlr13)
272 # quadratic term
273
274 mlr14 <- lm( log(Interactions + 1) ~ Category + Type + Paid +
275            Weekday + Hour
276            + LikeTotal + I(LikeTotal^2) + I(LikeTotal^3), data =
277            train)
278 summary(mlr14)
279 # cubic term
280
281 AIC(mlr1); AIC(mlr12); AIC(mlr13); AIC(mlr14);
282 # 5953.021; 1221.093; 1222.867; 1222.444;
283 # AIC of log transform model works better w/o nonlinear terms

```

```

280 # mlr12 is the best
281
282 # MLR part 2 =====
283 mlr2 <- lm(Interactions ~ Category + Type + Paid + Weekday + Hour
284           + LikeTotal + Impressions, data = train)
285 summary(mlr2)
286 # impressions is very valuable
287
288 mlr22 <- lm( log(Interactions + 1) ~ Category + Type + Paid +
289            Weekday + Hour
290            + LikeTotal + log(Impressions), data = train)
291 summary(mlr22)
292 # transform part 2
293
294 mlr23 <- lm( log(Interactions + 1) ~ Category + Type + Paid +
295            Weekday + Hour
296            + LikeTotal + I(LikeTotal^2) + log(Impressions), data =
297                train)
298 summary(mlr23)
299 # quadratic LikeTotal; non significant
300
301 mlr24 <- lm( log(Interactions + 1) ~ Category + Type + Paid +
302            Weekday + Hour
303            + LikeTotal + I(LikeTotal^2) + I(LikeTotal^3)
304            + log(Impressions), data = train)
305 summary(mlr24)
306 # cubic LikeTotal; non significant
307
308 mlr25 <- lm( log(Interactions + 1) ~ Category + Type + Paid +
309            Weekday + Hour
310            + LikeTotal + I(LikeTotal^2) + I(LikeTotal^3)
311            + log(Impressions) + + log(I(Impressions^2)), data =
312                train)
313 summary(mlr25)
314 # cubic log Impressions; na
315
316 mlr26 <- lm( log(Interactions + 1) ~ Category + Type + Paid +
317            Weekday + Hour
318            + LikeTotal + I(LikeTotal^2) + I(LikeTotal^3)
319            + log(Impressions) + LikeTotal*log(Impressions), data =
320                train)
321 summary(mlr26) # significant interaction
322
323 AIC(mlr2) # 5900.505
324 AIC(mlr22) # 944.728
325 AIC(mlr23) # 943.0069
326 AIC(mlr24) # 943.2385

```

```

319 AIC(mlr25) # 943.2385
320 AIC(mlr26) # 941.823
321 # pick mlr22; significant values unlike mlr1 for i=3:6 which are non
    significant
322
323 # LASSO/Ridge Regression ====
324 par(mfrow=c(1,1))
325
326 # best model for part 1:
327 mlr12 <- lm( log(Interactions + 1) ~ Category + Type + Paid +
    Weekday + Hour
328             + LikeTotal, data = train)
329 summary(mlr12)
330
331 # best model for part 2
332 mlr22 <- lm( log(Interactions + 1) ~ Category + Type + Paid +
    Weekday + Hour
333             + LikeTotal + log(Impressions), data = train)
334 summary(mlr22)
335 # using best model, will create based on this
336
337 # LASSO/RIDGE Model Selection =====
338 # we will need the glmnet R package:
339 library(glmnet)
340
341 # variables to use =====
342 # use categorical vars
343 Category;
344 Type;
345 Paid;
346 Weekday;
347 Hour;
348
349 # quantitative
350 LikeTotal;
351 ImpressionsTotal; # part 2, renamed as Impressions
352
353 # ====
354 X1 <- train$Category;
355 X2 <- train$Type;
356 X3 <- train$Paid;
357 X4 <- train$Weekday;
358 X5 <- train$Hour;
359 X6 <- train$LikeTotal
360 X7 <- log(train$Impressions) # part 2
361
362 # Category levels

```

```

363 X1_Product <- ifelse(X1 == "2", 1, 0)
364 X1_Inspiration <- ifelse(X1 == "3", 1, 0)
365
366 # Type levels
367 X2_Photo <- ifelse(X2 == "Photo", 1, 0)
368 X2_Status <- ifelse(X2 == "Status", 1, 0)
369 X2_Video <- ifelse(X2 == "Video", 1, 0)
370
371 # Weekday levels
372 X4_Mo <- ifelse(X4 == "2", 1, 0)
373 X4_Tu <- ifelse(X4 == "3", 1, 0)
374 X4_We <- ifelse(X4 == "4", 1, 0)
375 X4_Th <- ifelse(X4 == "5", 1, 0)
376 X4_Fr <- ifelse(X4 == "6", 1, 0)
377 X4_Sa <- ifelse(X4 == "7", 1, 0)
378
379 # Hour Levels
380 X5_morning <- ifelse(X5 >= 6 & X5 <= 11, 1, 0)
381 X5_afternoon <- ifelse(X5 >= 12 & X5 <= 17, 1, 0)
382 X5_evening <- ifelse(X5 >= 18 & X5 <= 23, 1, 0)
383
384 # XPi where i = 1 (part 1), 2 (part 2)
385 XP1 <- cbind( X1_Product, X1_Inspiration,
386               X2_Photo, X2_Status, X2_Video, X3,
387               X4_Mo, X4_Tu, X4_We, X4_Th, X4_Fr, X4_Sa,
388               X5_morning, X5_afternoon, X5_evening,
389               X6)
390
391 XP2 <- cbind( X1_Product, X1_Inspiration,
392               X2_Photo, X2_Status, X2_Video, X3,
393               X4_Mo, X4_Tu, X4_We, X4_Th, X4_Fr, X4_Sa,
394               X5_morning, X5_afternoon, X5_evening,
395               X6, X7)
396 YP <- log(train$Interactions + 1)
397
398 # select a "grid" of candidate lambda values (penalty sizes):
399 lambdas <- 2^seq(-10,10,length.out=100)
400
401 # use cross validation to obtain error estimates for lambda in grid:
402 # (using alpha=1, LASSO; alpha=0 Ridge )
403 lasso1 <- cv.glmnet(XP1,YP,lambda=lambdas,alpha=1)
404 lasso2 <- cv.glmnet(XP2,YP,lambda=lambdas,alpha=1)
405 ridge1 <- cv.glmnet(XP1,YP,lambda=lambdas,alpha=0)
406 ridge2 <- cv.glmnet(XP2,YP,lambda=lambdas,alpha=0)
407
408 # components:
409 names(lasso1); names(lasso2)

```

```

410 names(ridge1); names(ridge2)
411
412 # Plot      vs error
413
414 par(mfrow=c(1,2))
415 # Lasso Part 1
416 plot(lasso1$lambda,lasso1$cvm,type="l",log="x",
417       xlab = "Log(Lambda)", ylab = "Cross-Validated Error",
418       main = "Lasso CV Null Model")
419 abline(v=lasso1$lambda.min,col="orange")
420
421 MSE_l1 <- summary(lm(YP ~ XP1))$sigma^2
422 abline(h = MSE_l1, col = "purple", lwd = 2)
423 abline(h = var(YP), col = "darkred", lwd = 2)
424
425 # best lambda
426 lasso1_best <- lasso1$lambda.min
427 lasso1_best # 0.04282347
428
429 # final lasso1
430 final_lasso1 <- glmnet(XP1, YP,"gaussian", alpha = 1, lambda=lasso1_
  best)
431 yhat_lasso1 <- predict(final_lasso1, newx = XP1)
432 rmse1_lasso1_train <- sqrt(mean((YP - yhat_lasso1)^2))
433 rmse1_lasso1_train # 1.069372
434
435 # Ridge Part 1
436 plot(ridge1$lambda,ridge1$cvm,type="l",log="x",
437       xlab = "Log(Lambda)", ylab = "Cross-Validated Error",
438       main = "Ridge CV Null Model")
439 abline(v=ridge1$lambda.min,col="red")
440
441 MSE_R1 <- summary(lm(YP ~ XP1))$sigma^2
442 abline(h = MSE_R1, col = "purple", lwd = 2)
443 abline(h = var(YP), col = "darkred", lwd = 2)
444
445 # best lambda
446 ridge1_best <- ridge1$lambda.min
447 ridge1_best # 0.114125
448
449 # final ridge1
450 final_ridge1 <- glmnet(XP1, YP,"gaussian", alpha=0, lambda=ridge1_
  best)
451 yhat_ridge1 <- predict(final_ridge1, newx = XP1)
452 rmse1_ridge1_train <- sqrt(mean((YP - yhat_ridge1)^2))
453 rmse1_ridge1_train # 1.060108
454

```

```

455 # Lasso Part 2
456 plot(lasso2$lambda, lasso2$cvm, type="l", log="x",
457       xlab = "Log(Lambda)", ylab = "Cross-Validated Error",
458       main = "Lasso CV Full Model")
459 abline(v=lasso2$lambda.min, col="orange")
460
461 MSE_l2 <- summary(lm(YP ~ XP2))$sigma^2
462 abline(h = MSE_l2, col = "purple", lwd = 2)
463 abline(h = var(YP), col = "darkred", lwd = 2)
464
465 # best lambda
466 lasso2_best <- lasso2$lambda.min
467 lasso2_best # 0.0121438
468
469 # final lasso2
470 final_lasso2 <- glmnet(XP2, YP, "gaussian", alpha=1, lambda=lasso2_
  best)
471 yhat_lasso2 <- predict(final_lasso2, newx = XP2)
472 rmse2_lasso2_train <- sqrt(mean((YP - yhat_lasso2)^2))
473 rmse2_lasso2_train # 0.7267112
474
475 # Ridge Part 2
476 plot(ridge2$lambda, ridge2$cvm, type="l", log="x",
477       xlab = "Log(Lambda)", ylab = "Cross-Validated Error",
478       main = "Ridge CV Full Model")
479 abline(v=ridge2$lambda.min, col="red")
480
481 MSE_R2 <- summary(lm(YP ~ XP2))$sigma^2
482 abline(h = MSE_R2, col = "purple", lwd = 2)
483 abline(h = var(YP), col = "darkred", lwd = 2)
484
485 # best lambda
486 ridge2_best <- ridge2$lambda.min
487 ridge2_best # 0.02813451
488
489 # final ridge2
490 final_ridge2 <- glmnet(XP2, YP, "gaussian", alpha=0, lambda=ridge2_
  best)
491 yhat_ridge2 <- predict(final_ridge2, newx = XP2)
492 rmse2_ridge2_train <- sqrt(mean((YP - yhat_ridge2)^2))
493 rmse2_ridge2_train # 0.725468
494
495 # coefficients and beta comparisons for best versions of lasso and
  ridge
496 lasso1$beta
497 coef(final_lasso1);
498 coef(mlr12)

```

```

499 rmse1_lasso1_train # 1.069372
500
501 ridge1$beta
502 coef(final_ridge1)
503 coef(mlr12)
504 rmse1_ridge1_train # 1.060108
505
506 lasso2$beta
507 coef(final_lasso2);
508 coef(mlr22)
509 rmse2_lasso2_train # 0.7267112
510
511 ridge2$beta
512 coef(final_ridge2)
513 coef(mlr22)
514 rmse2_ridge2_train # 0.725468
515
516 # STEP 6; CROSS VALIDATION  =====
517
518 # rows of training data
519 n_cv <- nrow(train)
520
521 # make vector to store predicted probabilities:
522 y_hat1_cv <- numeric(n_cv)
523 y_hat2_cv <- numeric(n_cv)
524
525 # set seed again:
526 set.seed(4990)
527
528 # get a random permutation of all observations:
529 perm <- sample(n_cv)
530
531 # number of folds:
532 k <- 10
533
534 # fold size:
535 n_fold <- n_cv/k
536
537 # rename models
538 mdl1 <- mlr12
539 mdl2 <- mlr22
540
541 # loop over my k=10 folds:
542 for (j in 1:k) {
543   # observations from current fold:
544   fold <- perm[1:n_fold + (j-1)*n_fold]
545

```

```

546 # fit on all other folds:
547 mlr1 <- lm mdl1, data = train, subset = -fold)
548 mlr2 <- lm mdl2, data = train, subset = -fold)
549
550 # predict on the fold:
551 y_hat1_cv[fold] <- predict(mlr1, newdata=train[fold,])
552 y_hat2_cv[fold] <- predict(mlr2, newdata=train[fold,])
553 }
554
555 # calculate a cross-validated root mean squared error:
556 rmse1_cv <- sqrt(mean(( log(train$Interactions + 1) - y_hat1_cv)^2))
557 rmse2_cv <- sqrt(mean(( log(train$Interactions + 1) - y_hat2_cv)^2))
558
559 # RMSE if we did NOT do any cross validation:
560 mlr1_fit <- lm(mlr12, data=train)
561 rmse1_no_cv <- sqrt(mean(residuals(mlr12)^2))
562
563 mlr2_fit <- lm(mlr22, data=train)
564 rmse2_no_cv <- sqrt(mean(residuals(mlr22)^2))
565
566 # compare the error estimates:
567 rmse1_cv # 1.119299
568 rmse1_no_cv # 1.08592
569 summary(mlr1)$sigma # 1.11662
570
571 rmse2_cv # 0.7906019
572 rmse2_no_cv # 0.7668005
573 summary(mlr2)$sigma # 0.7928469
574
575 # STEP 7; COMPARE ALL MODELS TO SEE WHICH IS BEST
576
577 # Part 1
578 summary(mlr12)
579 rmse1_cv # 1.119299
580 rmse1_no_cv # 1.08592
581 summary(mlr1)$sigma # 1.11662
582
583 lasso1_best # 0.04282347
584 final_lasso1
585 rmse1_lasso1_train # 1.069372
586
587 ridge1_best # 0.114125
588 final_ridge1
589 rmse1_ridge1_train # 1.060108
590
591 # Part 2
592 summary(mlr22)

```

```

593 rmse2_cv # 0.7906019
594 rmse2_no_cv # 0.7668005
595 summary(mlr2)$sigma # 0.7928469
596
597 lasso2_best # 0.0121438
598 final_lasso2
599 rmse2_lasso2_train # 0.7267112
600
601 ridge2_best # 0.02813451
602 final_ridge2
603 rmse2_ridge2_train # 0.725468
604
605 # Step 8; test data
606
607 # MLR
608 mdl1 <- mlr12 # Part 1 best
609 mdl2 <- mlr22 # Part 2 best
610
611 # Part 1
612 # best model for part 1:
613 yhat_mlr1 <- predict(mlr12, newdata = test)
614 rmse_mlr1_test <- sqrt(mean((log(test$Interactions + 1) - yhat_mlr1)
615   ^2))
616 rmse_mlr1_test # 1.055665
617
618 # recreate X variables for test
619 X1_t <- test$Category;
620 X2_t <- test$Type;
621 X3_t <- test$Paid;
622 X4_t <- test$Weekday;
623 X5_t <- test$Hour;
624 X6_t <- test$LikeTotal
625 X7_t <- log(test$Impressions) # part 2
626
627 # categorical fix
628 # Category levels
629 X1_t_Product <- ifelse(X1_t == "2", 1, 0)
630 X1_t_Inspiration <- ifelse(X1_t == "3", 1, 0)
631
632 # Type levels
633 X2_t_Photo <- ifelse(X2_t == "Photo", 1, 0)
634 X2_t_Status <- ifelse(X2_t == "Status", 1, 0)
635 X2_t_Video <- ifelse(X2_t == "Video", 1, 0)
636
637 # Weekday levels
638 X4_t_Mo <- ifelse(X4_t == "2", 1, 0)
639 X4_t_Tu <- ifelse(X4_t == "3", 1, 0)

```

```

639 X4_t_We <- ifelse(X4_t == "4", 1, 0)
640 X4_t_Th <- ifelse(X4_t == "5", 1, 0)
641 X4_t_Fr <- ifelse(X4_t == "6", 1, 0)
642 X4_t_Sa <- ifelse(X4_t == "7", 1, 0)
643
644 # Hour Levels
645 X5_t_morning <- ifelse(X5_t >= 6 & X5_t <= 11, 1, 0)
646 X5_t_afternoon <- ifelse(X5_t >= 12 & X5_t <= 17, 1, 0)
647 X5_t_evening <- ifelse(X5_t >= 18 & X5_t <= 23, 1, 0)
648
649 # XPi_t where i = 1 (part 1), 2 (part 2)
650 XP1_t <- cbind( X1_t_Product, X1_t_Inspiration,
651                X2_t_Photo, X2_t_Status, X2_t_Video, X3_t,
652                X4_t_Mo, X4_t_Tu, X4_t_We, X4_t_Th, X4_t_Fr, X4_t_Sa
653                ,
654                X5_t_morning, X5_t_afternoon, X5_t_evening,
655                X6_t)
656 XP2_t <- cbind( X1_t_Product, X1_t_Inspiration,
657                X2_t_Photo, X2_t_Status, X2_t_Video, X3_t,
658                X4_t_Mo, X4_t_Tu, X4_t_We, X4_t_Th, X4_t_Fr, X4_t_Sa
659                ,
660                X5_t_morning, X5_t_afternoon, X5_t_evening,
661                X6_t, X7_t)
662
663 # LASSO
664 yhat_lasso1 <- predict(final_lasso1, newx = XP1_t)
665 # newx instead of newdata because it gave error in r
666
667 # Ridge
668 yhat_ridge1 <- predict(final_ridge1, newx = XP1_t)
669 # newx instead of newdata because it gave error in r
670
671 # TRUE VALUES (REUSE IN PART 2)
672 y_true <- log(test$Interactions + 1)
673
674 # RMSE
675 rmse_lasso1_test <- sqrt(mean((y_true - yhat_lasso1)^2))
676 rmse_ridge1_test <- sqrt(mean((y_true - yhat_ridge1)^2))
677
678 rmse_lasso1_test # 1.052764
679 rmse_ridge1_test # 1.053205
680
681 # best model for part 2
682 yhat_mlr2 <- predict(mlr22, newdata = test)
683 rmse_mlr2_test <- sqrt(mean((log(test$Interactions + 1) - yhat_mlr2)
684                             ^2))

```

```

683 rmse_mlr2_test # 0.7446126
684
685 # LASSO
686 yhat_lasso2 <- predict(final_lasso2, newx = XP2_t)
687 # newx instead of newdata because it gave error in r
688
689 # Ridge
690 yhat_ridge2 <- predict(final_ridge2, newx = XP2_t)
691 # newx instead of newdata because it gave error in r
692
693 # RMSE
694 rmse_lasso2_test <- sqrt(mean((y_true - yhat_lasso2)^2))
695 rmse_ridge2_test <- sqrt(mean((y_true - yhat_ridge2)^2))
696
697 rmse_lasso2_test # 0.733347
698 rmse_ridge2_test # 0.7411516
699
700 # STEP 9 Compare RMSE values
701
702 # Part 1
703
704 #TRAIN
705 summary(mlr12)
706 rmse1_cv # 1.119299
707 rmse1_no_cv # 1.08592
708 summary(mlr1)$sigma # 1.11662
709
710 lasso1_best # 0.04282347
711 final_lasso1
712 rmse1_lasso1_train # 1.069372
713
714 ridge1_best # 0.114125
715 final_ridge1
716 rmse1_ridge1_train # 1.060108
717
718 #TEST
719 rmse_mlr1_test # 1.055665
720 rmse_lasso1_test # 1.052764
721 rmse_ridge1_test # 1.053205
722
723 # Part 2
724
725 #TRAIN
726 summary(mlr22)
727 rmse2_cv # 0.7906019
728 rmse2_no_cv # 0.7668005
729 summary(mlr2)$sigma # 0.7928469

```

```
730
731     lasso2_best # 0.0121438
732     final_lasso2
733     rmse2_lasso2_train # 0.7267112
734
735     ridge2_best # 0.02813451
736     final_ridge2
737     rmse2_ridge2_train # 0.725468
738
739     #TEST
740     rmse_mlr2_test # 0.7446126
741     rmse_lasso2_test # 0.733347
742     rmse_ridge2_test # 0.7411516
```